RAÚL GARCÍA LEMUS

Software Development Engineer

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SUMMARY

Software engineer with 6+ years of experience across the full software development lifecycle — requirements, design and architecture, implementation, testing, and deployment. Hands-on experience building cloud-connected applications on AWS (EC2, S3, Lambda) and Firebase, REST APIs and webhook-driven automation, and cross-platform mobile/desktop software (Python, C#, Java/Kotlin), combined with software and systems architecture work on large, safety-critical platforms at Ford Motor Company. Recently shipped full-stack web projects using React, TypeScript, and Node.js/Express, including a voice-enabled productivity app built for an AI hackathon. Strong grounding in object-oriented design and design patterns, comfortable driving code and design reviews, and effective operating in ambiguous, fast-moving, cross-functional environments.

TECHNICAL SKILLS

Programming Languages: Python, C#, Java (Android/Kotlin), JavaScript (ES6+), TypeScript, C/C++, SQL

Web & Frontend: React, HTML5, CSS3 (Flexbox/Grid, BEM), Tailwind CSS, responsive design, i18next, PWA (Workbox)

Backend & Data: Node.js/Express, REST APIs, webhooks, Supabase (PostgreSQL, Auth, RLS, Edge Functions), Firebase, MySQL, MongoDB, DynamoDB

Cloud & Automation: AWS (EC2, S3, Lambda, IAM), event-driven & serverless architecture, n8n workflow automation, Google Cloud Platform APIs (Gmail, Drive)

Software Engineering Practices: Full SDLC, Object-Oriented Design, Design Patterns, Agile/Scrum (SAFe), Git/GitHub, code reviews & design reviews

Additional: UML/SysML modeling, applied AI/LLM tooling (RAG, LangChain), AI-assisted development (Codex), cross-functional program coordination

PROFESSIONAL EXPERIENCE

Ford Motor Company — 2021 – 2025

Platform System Engineer – Enclosures | *Software & Systems Engineering*

- Owned software and system architecture definition for embedded platform features, translating business and safety requirements into technical specifications adopted across multiple concurrent programs.
- Designed and executed a simulation-first validation strategy that verified 50%+ of software functionality before hardware availability, reducing downstream integration defects and shortening release timelines.
- Led modeling and architecture-definition strategy (MBSE/UML), producing the majority of the technical architecture standard adopted for new platform proposals.
- Led design and code reviews for safety-critical software components, proactively identifying architecture risks (Software FMEA) and raising test coverage.
- Partnered daily with software, hardware, and program management teams across multiple concurrent programs to unblock delivery and align on ambiguous, evolving requirements.
- Analyzed large-scale vehicle telemetry data to surface failure patterns, informing predictive-maintenance features.

TASIC S.A. de C.V. — 2018 – 2021

Service and Development Engineer | *Full-Stack & Embedded Software*

- Designed and built a cloud-connected service platform on AWS (EC2, S3, Lambda) for remote equipment monitoring, including event-driven workflows and automated maintenance notifications — sole engineer, end-to-end ownership.
- Built native Android applications (Java/Kotlin) for field data capture, shipped to production and used daily by field technicians.
- Designed desktop application architecture and UI (C#, WPF/Windows Forms) applying object-oriented design and UML modeling, reducing recurring system faults.
- Optimized SQL queries and integrated MongoDB for backend data handling, improving report retrieval performance and dashboard responsiveness for management.
- Built a computer vision feature that improved detection accuracy by 95%, and real-time control software in C that cut system response time by 50%.

SELECTED PROJECTS

Recuérdame — AI Training Hackathon (Kanz), July 2026 | solo developer | soon published on Kanz's AI portfolio

- Built a web app for managing reminders, person-linked tasks, and shopping lists, with a multilingual voice assistant that operates on the authenticated user's own data and respects their selected language.
- Stack: React + TypeScript + Vite, Tailwind CSS, Node.js/Express API, Supabase (PostgreSQL, Auth, Row-Level Security, Edge Functions), ElevenLabs Agents for voice, TanStack Query, i18next, and a PWA (Workbox) build, deployed on Replit with GitHub version control.

Quiniela Malenka — github.com/Beetik/Quiniela-Malenka

- World Cup prediction-pool web app adapted from an original Android app; front end in HTML5/CSS3/JavaScript, Firebase backend, self-hosted on a personal AWS EC2 server.
- Designed and deployed two production n8n automation workflows (hosted on a subdomain of the same server): one pulling live scores, upcoming matches, and match metadata (scorers, venues, kickoff times); another sending automated confirmation emails via webhooks, using GCP-authorized access to Gmail and Drive.
- Owned the system architecture and integration design end-to-end; implementation was AI-assisted (Codex).

Around the U.S. — raulgarlem.github.io/web_project_around_es | github.com/Raulgarlem/web_project_around_es

- Interactive front-end app for managing a collection of favorite places: dynamic card creation/deletion, profile editing, like toggling, and full-size image modals — built with vanilla JavaScript (ES6+), DOM manipulation, and event handling, with a fully responsive CSS Grid/Flexbox layout (BEM).

CONTINUING EDUCATION

Full Stack Software Engineering Bootcamp | TripleTen | 2026 – Present (in progress)

- Currently completing coursework and hands-on projects in JavaScript (ES6+), DOM manipulation, and Node.js/Express for backend API development.

EDUCATION & CERTIFICATIONS

M.S. Electrical Engineering – Digital Signal Processing (in progress) | UNAM – Posgrado de Ingeniería
B.S. Mechatronics Engineering | UNAM – Facultad de Ingeniería | 2014 – 2019

- Object-Oriented Design & Design Patterns — Coursera, University of Alberta (2022–2023)
- SW Architecture and Design of Modern Large-Scale Systems — Udemy (2022)
- Clean Code — Clean Coders (2025)
- Certified SAFe 5 PO/PM — Scaled Agile Inc. (2022)

LANGUAGES

Spanish (Native) | English (Advanced) | French (Intermediate)